

## COVE POINT LNG PIER, MD, USA

WMO: 997781

Lat: 38.402N

Lon: 76.385W

Elev: 2

StdP: 101.30

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |            |            |            |            | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD<br>to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|------------|------------|------------|------------|-----------------------|-------------|-------------|-------------|--------------------------|------------|--------------|
|               |             |             | 99.6%                         |            |            | 99%        |            |            | 0.4%                  |             | 1%          |             |                          |            |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB       | DP         | HR         | MCDB       | WS                    | MCDB        | WS          | MCDB        | MCWS                     | PCWD       |              |
| (a)<br>1      | (b)<br>-8.2 | (c)<br>-5.9 | (d)<br>N/A                    | (e)<br>N/A | (f)<br>N/A | (g)<br>N/A | (h)<br>N/A | (i)<br>N/A | (j)<br>15.4           | (k)<br>-3.1 | (l)<br>14.5 | (m)<br>-1.9 | (n)<br>7.8               | (o)<br>330 | (p)<br>0.570 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |            |             |            |             |            | Evaporation WB/MCDB |            |            |            |            |            | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|------------|-------------|------------|-------------|------------|---------------------|------------|------------|------------|------------|------------|----------------------|------------|
|               |                        | 0.4%            |            | 1%          |            | 2%          |            | 0.4%                |            | 1%         |            | 2%         |            |                      |            |
|               |                        | DB              | MCWB       | DB          | MCWB       | DB          | MCWB       | WB                  | MCDB       | WB         | MCDB       | WB         | MCDB       | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>5.1             | (c)<br>31.5     | (d)<br>N/A | (e)<br>30.1 | (f)<br>N/A | (g)<br>28.9 | (h)<br>N/A | (i)<br>N/A          | (j)<br>N/A | (k)<br>N/A | (l)<br>N/A | (m)<br>N/A | (n)<br>N/A | (o)<br>4.3           | (p)<br>220 |

| Dehumidification DP/MCDB and HR |     |      |     |     |      |     |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-----|------|-----|-----|------|-----|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |     |      | 1%  |     |      | 2%  |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR  | MCDB | DP  | HR  | MCDB | DP  | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b) | (c)  | (d) | (e) | (f)  | (g) | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| N/A                             | N/A | N/A  | N/A | N/A | N/A  | N/A | N/A | N/A  | N/A           | N/A  | N/A  | N/A  | N/A  | N/A  | N/A               |

## Extreme Annual Design Conditions

| Extreme Annual WS |          |          | Extreme Annual Temperature |           |                    |          | n-Year Return Period Values of Extreme Temperature |           |            |           |            |           |            |           |          |          |
|-------------------|----------|----------|----------------------------|-----------|--------------------|----------|----------------------------------------------------|-----------|------------|-----------|------------|-----------|------------|-----------|----------|----------|
|                   |          |          | Mean                       |           | Standard Deviation |          | n=5 years                                          |           | n=10 years |           | n=20 years |           | n=50 years |           |          |          |
| 1%                | 2.5%     | 5%       | Min                        | Max       | Min                | Max      | Min                                                | Max       | Min        | Max       | Min        | Max       | Min        | Max       |          |          |
| (a) 13.7          | (b) 11.9 | (c) 10.7 | (d) DB                     | (e) -10.0 | (f) 35.1           | (g) 3.1  | (h) 1.6                                            | (i) -12.2 | (j) 36.2   | (k) -14.0 | (l) 37.1   | (m) -15.7 | (n) 38.0   | (o) -17.9 | (p) 39.2 | (q) N/A  |
| (4) 13.7          | (5) 11.9 | (6) 10.7 | (7) WB                     | (8) N/A   | (9) N/A            | (10) N/A | (11) N/A                                           | (12) N/A  | (13) N/A   | (14) N/A  | (15) N/A   | (16) N/A  | (17) N/A   | (18) N/A  | (19) N/A | (20) N/A |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 14.5   | 2.3   | 3.3   | 6.9   | 13.4  | 18.9  | 23.7  | 26.0  | 25.0  | 22.0  | 16.3  | 9.6   | 6.5   |
|                                                                        | DBStd       | 9.12   | 5.07  | 4.93  | 4.56  | 4.25  | 3.62  | 2.85  | 2.22  | 1.97  | 2.72  | 3.90  | 3.98  | 4.49  |
|                                                                        | HDD10.0     | 758    | 243   | 194   | 120   | 14    | 0     | 0     | 0     | 0     | 0     | 3     | 54    | 129   |
|                                                                        | HDD18.3     | 2186   | 496   | 420   | 356   | 158   | 38    | 2     | 0     | 0     | 3     | 85    | 261   | 368   |
|                                                                        | CDD10.0     | 2417   | 6     | 7     | 22    | 116   | 275   | 411   | 494   | 464   | 359   | 199   | 43    | 19    |
|                                                                        | CDD18.3     | 803    | 0     | 0     | 0     | 9     | 55    | 163   | 236   | 206   | 112   | 22    | 0     | 0     |
|                                                                        | CDH23.3     | 5259   | 0     | 1     | 2     | 59    | 193   | 987   | 2028  | 1465  | 476   | 48    | 0     | 0     |
|                                                                        | CDH26.7     | 1087   | 0     | 0     | 0     | 12    | 35    | 209   | 526   | 252   | 51    | 3     | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 5.2    | 5.5   | 5.6   | 5.8   | 5.6   | 5.1   | 4.8   | 4.4   | 4.4   | 5.0   | 5.6   | 5.8   | 5.2   |
| (15) Precipitation                                                     | PrecAvg     | 1102   | 90    | 85    | 101   | 79    | 103   | 91    | 105   | 95    | 103   | 79    | 80    | 86    |
|                                                                        | PrecMax     | 1492   | 175   | 168   | 263   | 208   | 279   | 189   | 469   | 236   | 238   | 168   | 176   | 191   |
|                                                                        | PrecMin     | 812    | 12    | 8     | 23    | 11    | 14    | 23    | 21    | 20    | 2     | 0     | 9     | 12    |
|                                                                        | PrecStd     | 204    | 41    | 39    | 56    | 43    | 56    | 48    | 82    | 62    | 60    | 46    | 44    | 50    |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 19.0  | 20.6  | 21.9  | 28.3  | 29.9  | 32.8  | 34.7  | 32.5  | 30.3  | 26.9  | 20.9  | 20.9  |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 2%          | DB     | 15.4  | 16.0  | 18.8  | 24.9  | 27.3  | 30.3  | 32.1  | 30.2  | 28.1  | 24.5  | 18.6  | 17.7  |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 5%          | DB     | 11.6  | 12.5  | 15.7  | 21.8  | 25.3  | 28.9  | 30.5  | 28.9  | 26.8  | 23.2  | 16.6  | 14.8  |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 10%         | DB     | 8.9   | 9.1   | 13.1  | 19.1  | 23.8  | 27.7  | 29.2  | 28.0  | 25.9  | 21.7  | 15.2  | 12.6  |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 2%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 5%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 10%         | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 5.9   | 6.2   | 6.0   | 6.7   | 6.0   | 5.3   | 5.1   | 4.7   | 4.5   | 5.2   | 5.3   | 5.3   |
|                                                                        |             | MCDBR  | 10.4  | 11.5  | 10.8  | 10.6  | 7.8   | 6.9   | 6.8   | 6.0   | 5.3   | 6.5   | 7.2   | 8.8   |
|                                                                        |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        | 5% WB       | MCDBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                        |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.311 | 0.320 | 0.355 | 0.395 | 0.436 | 0.487 | 0.517 | 0.508 | 0.412 | 0.360 | 0.326 | 0.312 |
|                                                                        | taud        |        | 2.505 | 2.474 | 2.422 | 2.327 | 2.243 | 2.150 | 2.079 | 2.088 | 2.334 | 2.481 | 2.536 | 2.543 |
|                                                                        | Ebn at Noon |        | 871   | 910   | 905   | 884   | 849   | 804   | 775   | 771   | 834   | 851   | 847   | 846   |
|                                                                        | Edh at Noon |        | 78    | 91    | 107   | 124   | 138   | 151   | 161   | 156   | 115   | 90    | 76    | 71    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 2.19  | 3.02  | 4.03  | 5.09  | 5.79  | 6.37  | 6.19  | 5.54  | 4.48  | 3.40  | 2.48  | 1.93  |
|                                                                        | RadStd      |        | 0.17  | 0.32  | 0.34  | 0.36  | 0.66  | 0.48  | 0.30  | 0.31  | 0.46  | 0.37  | 0.21  | 0.17  |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (82 neighbors) | +0.45 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +143    | +88     |

Nomenclature: See separate page